

Jacob Otero

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EDUCATION

The University of Alabama

Bachelor of Science, Computer Science

Relevant Coursework: Data Structures & Algorithms, Database Management, Software Engineering

Tuscaloosa, AL

Dec 2027

GPA: 3.45/4.00

WORK EXPERIENCE

Mercedes-Benz U.S. International

Software Engineering Co-op - GSP (Global Service & Parts)

Tuscaloosa, AL

May 2026 - Aug 2026

- Engineered end-to-end development of a compliance workflow application used by 18 different business units, from initial design through deployment, cutting average review/approval time from 10 hours/week to 2 hours by working directly with compliance staff to recreate the process
- Led a company-wide initiative to make an internal department tool available to international teams, coordinating departmental requirements and reducing app development time by 14 days per app

Mercedes-Benz U.S. International

Software Engineering Co-op - Battery Plant

Tuscaloosa, AL

Aug 2025 - Dec 2025

- Identified a gap in real-time production tracking and autonomously drafted, constructed, and launched a full-stack solution now utilized by 100s of engineers/operators across the plant, replacing a manual procedure that took 30 minutes/8 hours
- Created and prototyped 15 custom machine parts using CAD modeling and 3D printing, decreasing part turnaround time from 2 hours to 10 minutes

Mercedes-Benz U.S. International

Software Engineering Co-op - GSP (Global Service & Parts)

Tuscaloosa, AL

Jan 2025 - May 2025

- Partnered with the Dangerous Goods department to automate manual documentation steps, cutting errors/audit flags by 40% across global trade operations
- Digitalized a recurring administrative task that previously took 4 hours/week, saving the department roughly 16 hours per month

PROJECTS

DonorTrack

Personal Project, 2026

- Deployed a full-stack donor management platform (React/TypeScript, Node.js) giving nonprofits real-time dashboards to track donations and campaign performance, and deployed it on production AWS infrastructure (S3, CloudFront, Lambda, CI/CD) as a free, always-available tool for organizations

DG Site Review App

MBUSI, 2026

- Designed and assembled a full compliance review method in Power Apps/Power Automate, enabling employees to propose edits to dangerous goods compliance records with side-by-side diff comparisons, role-based approve/deny logic, and automated email notifications across distinct stages

AGV Fault Tracker

MBUSI, 2025

- Built a full-stack fault-tracking web app from scratch, accessible via QR codes mounted on AGVs for on-the-spot access from company phones; developed a JavaScript/HTML/CSS frontend, a Python/Flask REST API backend, and a MongoDB database to store and manage fault records, then packaged and deployed the completed application to the company's internal server

INVOLVEMENT

Association for Computing Machinery (ACM)

Software Development Member

The University of Alabama, AL

Aug 2024 - Present

- Collaborate on team coding challenges and build small applications to strengthen problem-solving and teamwork

Theta Tau (Engineering Fraternity)

Member

The University of Alabama, AL

Aug 2024 - Present

- Participate in professional engineering development workshops and networking events to gain industry insights and career guidance

SKILLS

- Languages:** Python, C++, Java, JavaScript, C#, SQL, Power Fx, HTML/CSS, MATLAB, VHDL
- Frameworks & Tools:** AWS, Docker, CI/CD, React.js, Node.js, Flask, MongoDB, Git, Scrum, Linux, JSON
- AI & Machine Learning:** Agents, RAG, LLM integration, MCP, REST APIs, TensorFlow, LangChain